

Multi-Spectral Vision System for Quality Inspection of In-Coupler Gratings in AR Displays

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Abstract—Diffraction in-coupler (IC) gratings are critical components in augmented-reality (AR) waveguides, yet their nanoscale structure makes them highly sensitive to fabrication defects that degrade optical efficiency and display quality. This work presents an automated hyperspectral measurement system for IC grating quality control, designed to convert customer requirements into three quantitative Critical-to-Quality (CTQ) metrics: Spectral Root Mean Square Error (RMSE), Spectral Angle Mapper (SAM), and spatial Uniformity. A complete preprocessing pipeline is developed, including reflectance computation, fiducial-based homography alignment, and circle-based IC region extraction.

Phase-1 evaluation on 14 representative defect samples demonstrates that the three CTQs reliably separate all defect categories — Periphery, Mask, Spot, Shape, and Homogeneity — based on their spectral-spatial signatures. Phase-2 implements a full Gage Repeatability and Reproducibility (Gage R&R) study using 12 representative samples measured by four operators across two trials. All three CTQs exhibit low measurement-system variation ($GR\&R < 10\%$), confirming suitability for industrial deployment.

The proposed system provides objective, repeatable, and interpretable defect quantification for diffractive gratings, enabling improved process control and production yield in next-generation AR waveguide manufacturing.

Index Terms—Hyperspectral Imaging, AR Waveguides, Diffractive Gratings, Critical-to-Quality Metrics, Spectral Angle Mapper, Gage R&R, Defect Inspection

I. INTRODUCTION

Augmented reality (AR) waveguide displays represent a transformative technology for immersive human-digital interaction, enabling seamless overlay of virtual content onto the real world through compact and lightweight optical systems. At the core of these displays are diffractive in-coupler (IC) gratings — nanoscale periodic structures that efficiently couple collimated light from micro-projectors into thin waveguides via controlled diffraction, ensuring high-efficiency light propagation and wide field-of-view (FOV) performance [1]. However, high-volume manufacturing of these gratings introduces defects such as periphery rings, global mask shifts, localized spots, shape distortions, and sectoral homogeneity variations, all of which degrade optical efficiency and reduce overall display yield [1].

This work was conducted in collaboration with Dispelix Oy, Joensuu, Finland.

The industrial challenge arises from the limitations of manual visual inspection for IC-grating quality control. These defects, often subtle and spatially variant, require a numerical evaluation using 2–3 Critical-to-Quality (CTQ) parameters capable of distinguishing true process-induced variations from measurement noise. Background research identifies five primary defect categories — Periphery, Mask, Spot, Shape, and Homogeneity — each exhibiting spectral-spatial signatures that challenge traditional RGB imaging due to their nanoscale periodicity and wavelength-dependent diffraction properties [1].

Recent literature demonstrates that hyperspectral imaging (HSI) has become a robust, non-destructive technique for defect metrology in diffractive optical elements and waveguides, outperforming conventional machine-vision systems by capturing continuous spectral data cubes for material-specific anomaly detection [2]–[4]. Classical algorithms such as Gabor filtering have been used in surface-defect extraction for optical gratings [5]. The Spectral Angle Mapper (SAM), a widely used HSI similarity metric, quantifies angular distance between pixel spectra and reference signatures to detect global inconsistencies while remaining insensitive to illumination variations [6], [7]. Complementary methods, including Fisher’s Linear Discriminant Analysis (FLD) applied to line-scan HSI, address dimensionality challenges in manufacturing-based defect classification [3]. Furthermore, Gage R&R analysis remains critical for validating measurement-system reliability in industrial metrology, with total Gage R&R values below 10% generally considered acceptable for production deployment [8].

The primary objective of this work is to develop a reliable, fully automated hyperspectral measurement system for quality control of diffractive in-coupler gratings in AR waveguide displays. The system provides a robust numerical assessment of fabrication-induced defects independent of manufacturing-process variation and measurement noise. This is achieved through a Voice-of-Customer (VoC)-to-Critical-to-Quality (CTQ) framework that defines three rigorously validated metrics:

- **CTQ1 (RMSE)** — Root Mean Square Error (RMSE) between measured and reference spectral reflectance curves, capturing global intensity deviations associated with mask shifts and periphery defects;

- **CTQ2 (SAM)** — Spectral Angle Mapper (SAM), measuring angular separation between spectral vectors to detect distortions in the diffraction response while remaining insensitive to illumination differences;
- **CTQ3 (Uniformity)** — Sector-wise statistical analysis (robust coefficient of variation and range ratio across eight radial sectors), sensitive to localized spots and regional inhomogeneity.

The proposed measurement pipeline integrates push-broom hyperspectral image acquisition with geometric alignment based on four-point homography. The three CTQs are computed independently, and a sample is accepted only when all CTQs meet their respective thresholds. All processing relies exclusively on classical computer-vision and multivariate statistical methods to ensure interpretability and robustness suitable for industrial integration. Extensive experimental validation on representative IC-grating samples spanning all defect categories, combined with an eight-trial multi-operator Gage R&R study, demonstrates that the total measurement-system variation satisfies stringent industrial acceptance criteria for high-volume manufacturing.

This work introduces a complete, industrially validated hyperspectral metrology platform that advances automated defect quantification for diffractive waveguide components. By providing objective, repeatable, and physically interpretable quality indices aligned with manufacturing requirements, the proposed system enables substantial improvements in yield, process control, and product consistency in next-generation AR display fabrication.

Objectives of the Study: The purpose of this study is to develop and validate an automated hyperspectral measurement system capable of detecting IC-grating defects using a set of three Critical-to-Quality (CTQ) metrics derived from customer requirements. Phase 1 focuses on CTQ selection and evaluation using fourteen defect samples representing the full spectrum of industrial variability. Phase 2 implements a full Gage Repeatability and Reproducibility (Gage R&R) study using twelve representative samples to assess measurement consistency across operators, trials, and defect types. The overarching goal is to determine whether the proposed CTQs — spectral RMSE, spectral angle mapper, and spectral uniformity — provide reliable and interpretable indicators of IC-grating quality suitable for industrial deployment.

II. MATERIALS AND METHODOLOGY

To determine whether a given in-coupler (IC) grating is defective, this work applies a template-matching verification method using hyperspectral imaging. This section describes the materials used, the dataset, the processing pipeline, and the Critical-to-Quality (CTQ) metrics derived from the Voice of the Customer (VoC).

A. Materials Used

The imaging system employed in this study is a multi-spectral camera based on a Liquid Crystal Tunable Filter (LCTF). Specifically, we used a CRI Nuance EX multispectral

CCD camera equipped with an LCTF module that enables wavelength-selective acquisition across 450–950 nm, covering both the visible (VIS) and near-infrared (NIR) regions with a 10 nm step size.

At each selected wavelength, the LCTF isolates a narrow spectral band, and the monochrome CCD sensor records a corresponding grayscale image. Sequential acquisition across all wavelengths produces a hyperspectral data cube suitable for spectral-reflectance analysis.

All measurements were conducted using a standardized D65 light-booth source in a controlled dark environment to minimize ambient-light interference. A printed IC grating reference, multiple defective IC gratings, and a white reference tile were measured for calibration and validation.

B. Dataset and Sample Description

The Phase 1 evaluation used fourteen IC grating samples representing all customer-defined defect types. High-priority defects include Periphery (Defects 1–2), Mask (Defects 3–5), and Homogeneity (Defects 11–12). Medium- and low-priority categories include Spot defects (Defects 6–8) and Shape variations such as Bigger/Smaller (Defects 9–10), Position shift (Defect 13), and Oval geometry (Defect 14). These samples were used to identify and validate suitable CTQs.

Phase 2 consisted of a Gage R&R study with twelve representative samples selected for measurement-system evaluation. These include Perfect samples (4, 13, 20), Mask defects (1, 2, 14), Spot defects (10, 12), Homogeneity defects (6, 15), and Shape anomalies (17: Position; 22: Horizontal Oval). This set provides adequate spectral and structural diversity to quantify repeatability and reproducibility across operators.

C. Data Processing and Region of Interest Detection

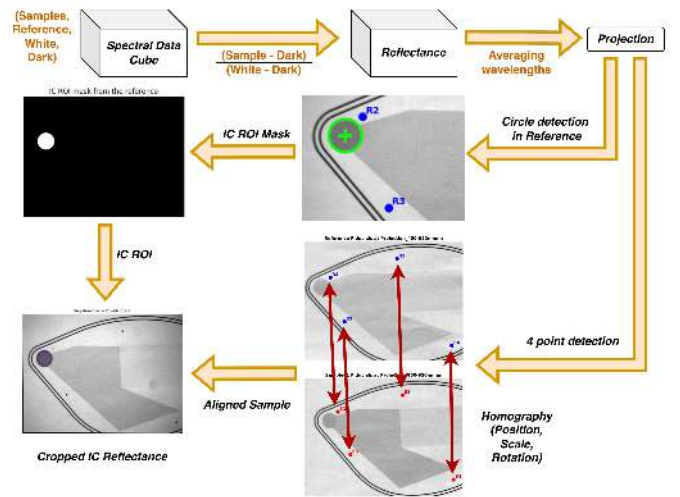


Fig. 1: Overview of the data-processing workflow, including 4-point detection, homography, IC ROI mask, and reflectance computation.

After measuring all samples along with white and dark references, spectral reflectance was computed pixel-wise using

$$R_s(X, Y, \lambda) = \frac{S_s(X, Y, \lambda) - S_d(X, Y, \lambda)}{S_w(X, Y, \lambda) - S_d(X, Y, \lambda)} \quad (1)$$

where: S_s = sample spectra, S_w = white reference, S_d = dark reference, R_s = reflectance.

A 2D projection image was generated by averaging the reflectance cube across wavelengths, enabling reliable geometric detection for template matching.

To ensure correct comparison between each sample and the reference, geometric alignment was performed by detecting four cross-shaped fiducial markers using thresholding, morphological filtering, and contour-based geometric selection. The detected markers were used to compute a four-point homography transformation that compensates for translation, rotation, and scale variations introduced during acquisition.

Simultaneously, the IC circular boundary was extracted using the Hough Circle Transform applied to a bilateral-filtered projection image. This provides a stable estimate of the IC center and radius. The combination of marker-based homography and circular ROI extraction ensures precise localization and alignment of the IC region for subsequent spectral comparison.

D. CTQs and Algorithms

After constructing the reflectance cube and aligning each sample to the reference, three CTQs derived from the VoC were defined to determine sample acceptability. Diffraction performance is highly sensitive to changes in grating material, height, and fill factor; thus, any structural non-uniformity produces measurable spectral deviations.

A sample passes inspection only if all three CTQs fall within their specified tolerance thresholds.

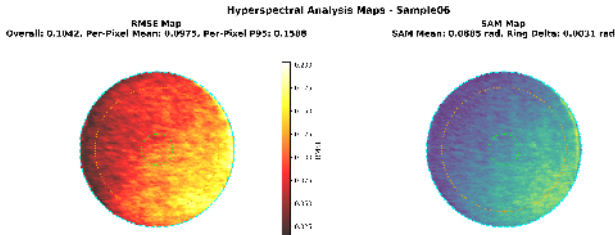


Fig. 2: RMSE map and SAM map

CTQ-1: Spectral Root Mean Square Error (SRMSE): SRMSE quantifies the magnitude difference between reference and test spectra across all wavelength bands. It is sensitive to brightness deviations and surface characteristics, such as periphery defects, localized spots, and shape distortions. The SRMSE is defined as

$$\text{SRMSE} = \sqrt{\frac{1}{N_p} \sum_{p=1}^{N_p} \left(\frac{1}{N_b} \sum_{b=1}^{N_b} (R_{pb}^{(\text{ref})} - R_{pb}^{(\text{test})})^2 \right)} \quad (2)$$

The tolerance requirement is $0 \leq \text{SRMSE} \leq 0.08$.

CTQ-2: Spectral Angle Mapper (SAM): SAM measures spectral-shape similarity by computing the angle between the reference and test spectral vectors. It is invariant to intensity changes and is particularly sensitive to color-shift defects such as mask variation. SAM is defined as

$$\text{SAM} = \arccos \left(\frac{R \cdot T}{\|R\| \|T\|} \right) \quad (3)$$

The tolerance requirement is $0 \leq \text{SAM} \leq 3.2^\circ$.

CTQ-3: Spectral Uniformity: Spectral uniformity evaluates spatial homogeneity within the IC region by partitioning it into angular sectors, computing the median reflectance in each sector, and quantifying variability. The normalized uniformity score is defined as

$$U = 1 - \left(\alpha \cdot \frac{\text{rCV}}{T_{\text{rCV}}} + (1 - \alpha) \cdot \frac{\text{RR}}{T_{\text{RR}}} \right) \quad (4)$$

with

$$\text{rCV} = \frac{\text{MAD}}{\text{median} + \varepsilon}, \quad \text{RR} = \frac{\text{max} - \text{min}}{\text{median} + \varepsilon}.$$

The tolerance requirement is $0.4 \leq U \leq 1$.

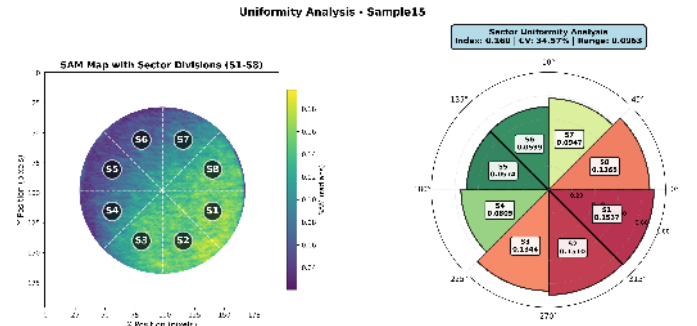


Fig. 3: Sector Uniformity

Final Classification Rule: A test sample is classified as *pass* if and only if:

$$\text{SRMSE} \leq 0.08, \quad \text{SAM} \leq 3.2^\circ, \quad 0.4 \leq U \leq 1.$$

Otherwise, the sample is considered *defective*.

III. EXPERIMENTS

A. Phase 1: Spectral Data Acquisition

Phase 1 focused on acquiring hyperspectral measurements from representative IC grating samples using the CRI/PerkinElmer Nuance EX multispectral imaging system. This LCTF-based system records narrow spectral bands sequentially to generate a full spectral cube (x, y, λ) spanning the 450–950 nm wavelength range with 10 nm steps. Figures 4–6 illustrate the camera, lighting configuration, and acquisition software interface.

The camera and D65 light source were arranged in a $0^\circ/45^\circ$ illumination–imaging geometry. Camera positioning was fixed for all measurements to maintain repeatability: the lower camera section was placed 475 mm above the ground, the

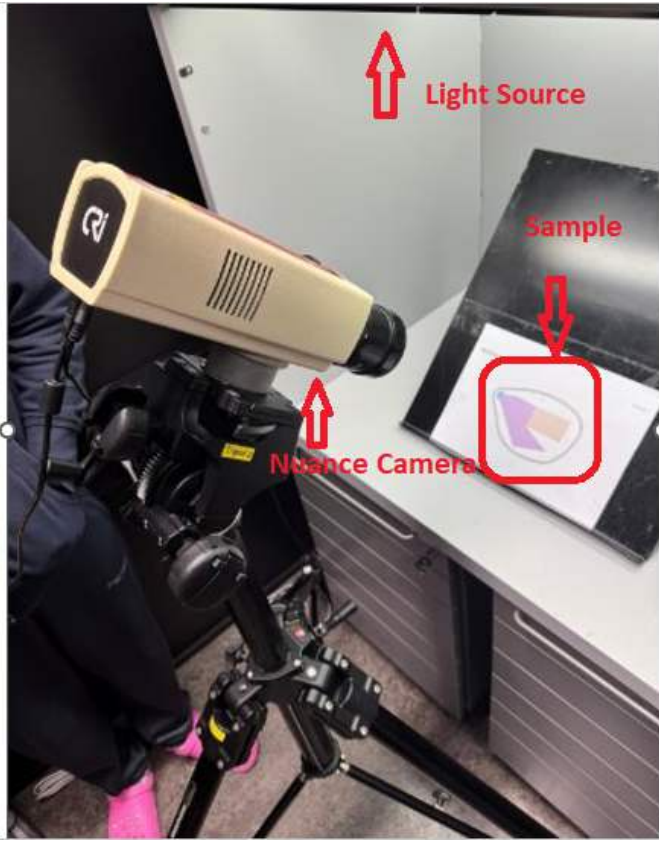


Fig. 4: Measurement setup naming for Phase 1.

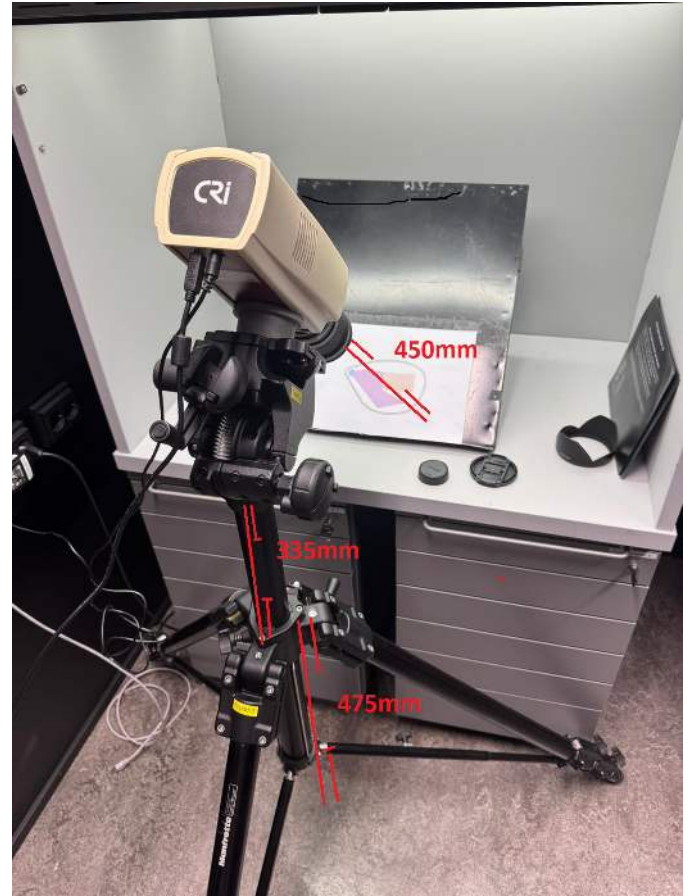


Fig. 6: Experimental setup for Phase 1 hyperspectral imaging.

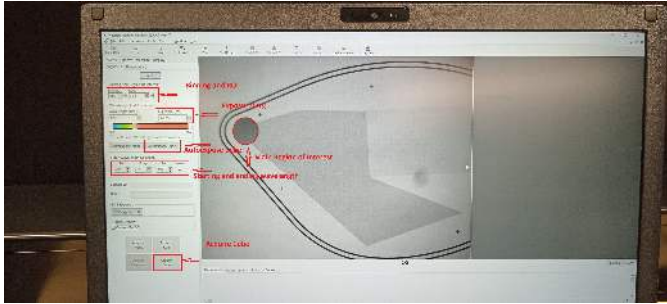


Fig. 5: CRI Nuance Camera Software used for Phase 1 acquisition.

upper section at 335 mm, and the lens-to-sample distance at 450 mm.

After positioning the camera, the CRI Nuance software (Fig. 5) was used to adjust focus and framing. Acquisition settings were configured as follows: full-frame ROI, 1×1 binning, 200 ms exposure time, and a spectral sweep from 450 nm to 950 nm with 10 nm increments. A white reference tile was acquired using the auto-expose cube function to ensure adequate illumination across the spectrum. Spectral cubes were then recorded for the printed reference IC, all fourteen defect samples, and finally a dark reference to capture sensor noise.

At the end of Phase 1, the following datasets were collected:

- 14 defect sample cubes of size (1040, 1392, 51),

- 1 defect-free reference cube of size (1040, 1392, 51),
- 1 white reference cube,
- 1 dark reference cube.

These data provided the basis for reflectance computation and CTQ evaluation in subsequent analysis.

B. Phase 2: Gage Repeatability and Reproducibility Study

Phase 2 consisted of a full Gage Repeatability and Reproducibility (Gage R&R) evaluation to quantify measurement variation introduced by equipment, operators, and sample-to-sample differences. Twelve IC grating samples spanning representative defect categories were used for this study.

TABLE I: Experimental Setup and Measurement Parameters for Gage R&R

Parameter	Description
Number of Samples	12 IC grating samples
Number of Operators	4
Number of Trials per Operator	2
Total Measurements	$12 \times 4 \times 2 = 96$ observations per CTQ
CTQs Evaluated	RMSE, SAM, Uniformity
Measurement Type	Operator-supervised

All measurements were collected in a controlled dark-room environment under identical D65 illumination, instrument settings, and imaging geometry. The Phase 2 setup (Fig. 7) used

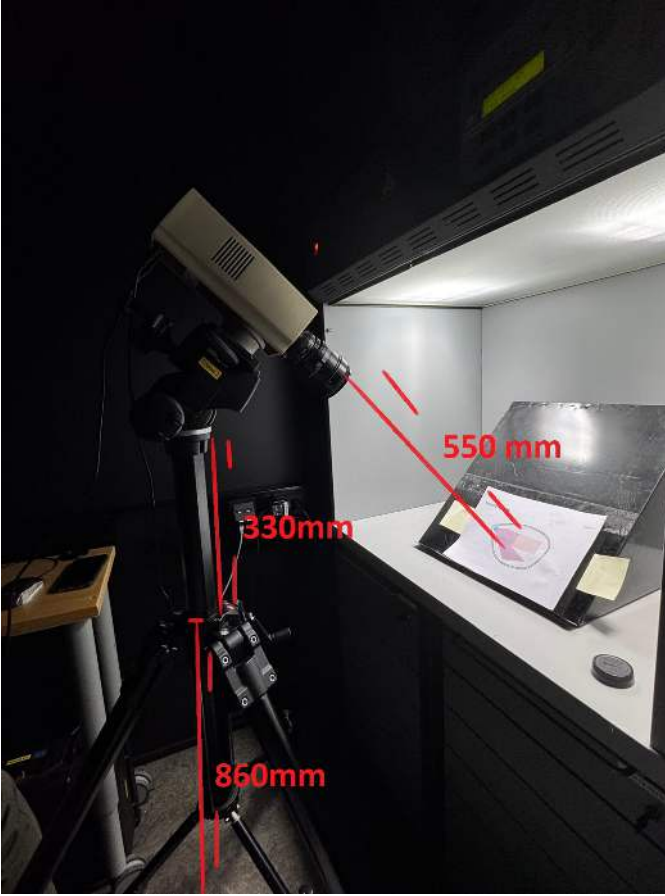


Fig. 7: Experimental setup for Phase 2 Gage R&R measurements.

the same $0^\circ/45^\circ$ configuration as Phase 1 but with updated positioning for improved operator accessibility: the lower camera section was fixed at 860 mm above the ground, the upper section at 330 mm, and the lens-to-sample distance at 550 mm.

Trial Procedure: Each of the four operators performed two complete measurement trials. For both Trial 1 and Trial 2, the following procedure was strictly enforced:

- Perform a fresh calibration, including white-reference correction.
- Acquire all twelve samples in the predefined sample order (see Appendix).
- Record a dark reference at the end of each trial.

Operators performed measurements sequentially to prevent cross-interference. Each operator's data were stored in operator-specific folders (e.g., `Operator1T1`, `Lue1T1`) and subsequently combined into the Group-C master directory for analysis.

This structured design ensured:

- 1) consistent calibration per operator
- 2) full independence between operators
- 3) valid estimation of repeatability (within-operator variation)

- 4) valid estimation of reproducibility (between-operator variation)

The resulting dataset of 96 observations per CTQ provided sufficient statistical power for a robust Gage R&R assessment aligned with industrial MSA guidelines.

IV. RESULTS

This section presents the experimental findings from both phases of the study.

- Phase 1 evaluates the discriminative power of the three Critical-to-Quality (CTQ) metrics using fourteen defect samples spanning all defect categories.
- Phase 2 analyzes the cross-operator consistency of the CTQs through a full Gage Repeatability and Reproducibility (Gage R&R) study on twelve representative samples.

A. Phase 1: CTQ Selection Using 14 Defect Samples

Phase 1 assessed the suitability of the three CTQs — Spectral Root Mean Square Error (SRMSE), Spectral Angle Mapper (SAM), and Spectral Uniformity — for detecting the full spectrum of defect modes encountered in IC grating manufacturing.

The dataset contained fourteen samples covering all customer-defined defect categories: high-priority Periphery (D1–D2), Mask (D3–D5), and Homogeneity (D11–D12), as well as medium- and low-priority Spot (D6–D8) and Shape variations (D9–D10: Bigger/Smaller; D13: Position shift; D14: Oval). Figure 8 illustrates the taxonomy, and Figure 9 summarizes the hyperspectral CTQ responses.

CTQ-1: SRMSE (Reflectance Magnitude Difference):

The SRMSE metric successfully captured brightness and magnitude related deviations between the reference and defect samples.

Periphery defects (D1–D2) exhibited low–moderate SRMSE values (≈ 0.06 – 0.09), whereas Mask (D3–D5) and Spot defects (D6–D8) generated higher deviations due to localized brightness changes. Shape-related distortions, particularly D13 (Position), produced the largest SRMSE (≈ 0.18), indicating substantial structural and spectral mismatch.

The threshold of $\text{SRMSE} \leq 0.08$ clearly separated acceptable samples from those with intensity-based or fill-factor-related anomalies.

CTQ-2: Spectral Angle Mapper (SAM): SAM demonstrated strong sensitivity to wavelength-dependent spectral-shape distortions.

Mask defects (D3–D5) produced some of the highest angular deviations (≈ 4 – 5°), consistent with spectral-shape distortion due to mask misalignment or height variations. Periphery defects remained close to the decision boundary, while D13 (Position) exhibited the largest angular deviation ($\approx 5.8^\circ$).

The SAM threshold ($\text{SAM} \leq 3.2^\circ$) therefore served as a reliable boundary for isolating defects that alter the spectral shape rather than magnitude.

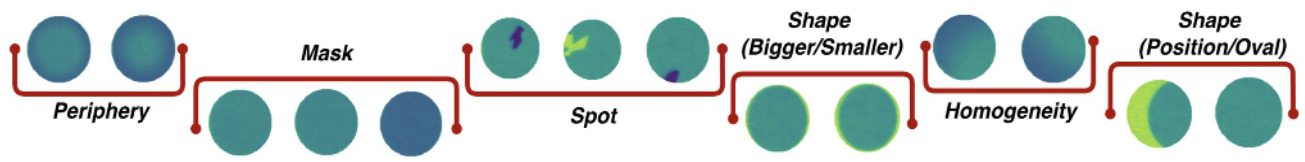


Fig. 8: Defect Types in Phase 1

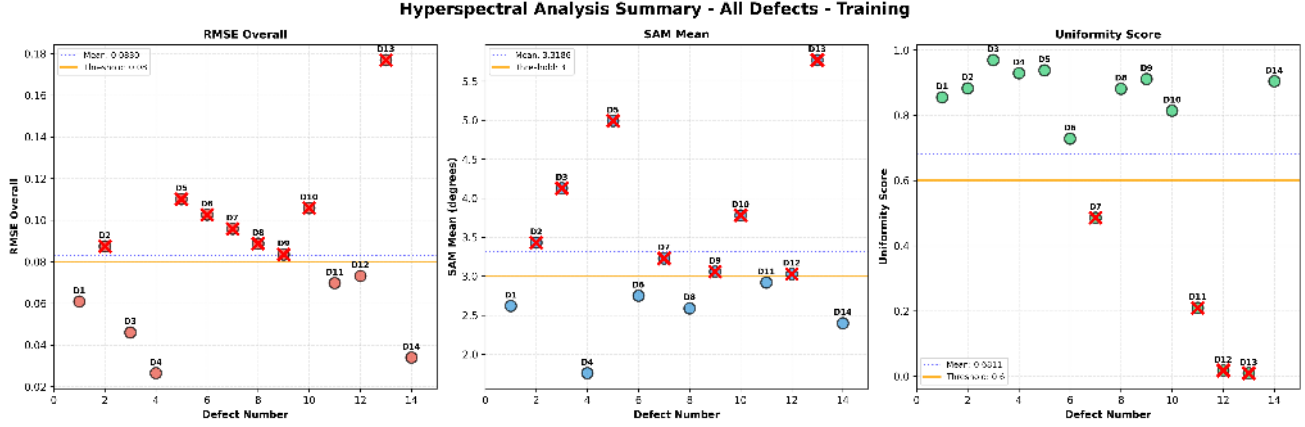


Fig. 9: Hyperspectral Analysis Summary across 12 samples in Phase 1

CTQ-3: Spectral Uniformity: The Uniformity metric showed strong discriminative capability for detecting spatial inhomogeneity.

Homogeneity defects (D11–D12) produced extremely low uniformity values (≈ 0 –0.2), indicating strong radial and angular intensity variation. Defect 13 (Position) and Spot-like anomalies also exhibited reduced uniformity. Conversely, Periphery and Mask defects maintained high uniformity (≈ 0.8 –0.95), reflecting their globally consistent structures.

The threshold of $U \geq 0.4$ cleanly separated spatially uniform samples from those with strong local gradients or directional inhomogeneity.

Summary — Phase 1: All three CTQs demonstrated clear separability among defect categories, and their predefined tolerance thresholds (SRMSE: 0.08, SAM: 3.2° , Uniformity: 0.4) were validated experimentally. Each CTQ captured distinct physical characteristics of the gratings, confirming that all three are required for robust defect detection.

B. Phase 2: Cross-Operator CTQ Measurement Consistency

Phase 2 evaluated the stability of the measurement system using a full crossed Gage R&R study on twelve representative IC grating samples. These included Perfect samples (4, 13, 20), Mask (1, 2, 14), Spot (10, 12), Homogeneity (6, 15), and Shape anomalies (17: Position; 22: Horizontal Oval). Each sample was measured independently by four operators across two trials, yielding 96 measurements per CTQ. Figures 10, 11, and 12 show the full operator distributions.

CTQ-1: SRMSE Consistency Across Operators: The SRMSE results exhibited tightly clustered distributions for

nearly all samples, demonstrating strong repeatability and reproducibility.

Perfect samples (4, 13, 20) consistently produced the smallest SRMSE values, well below the 0.08 threshold, confirming measurement stability for defect-free structures.

Mask, Spot, and Homogeneity defects exhibited moderately higher values, but with minimal operator-to-operator variance.

Shape defects (17 and 22) produced the highest SRMSE values due to geometric distortion, yet measurements remained consistent across all operators and trials. Overall, SRMSE showed excellent cross-operator consistency.

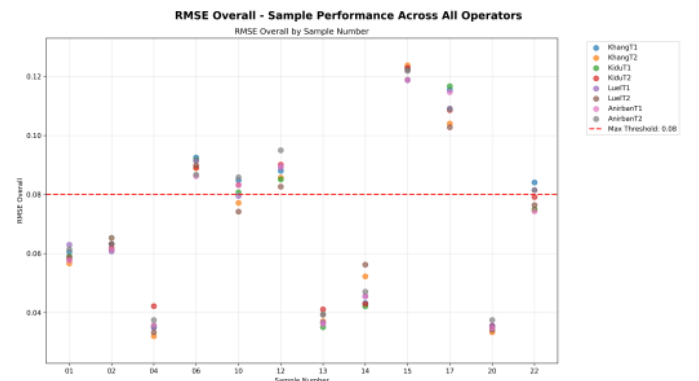


Fig. 10: SRMSE Across All Operators

CTQ-2: SAM Consistency Across Operators: SAM results were highly repeatable, with operator curves nearly overlapping for every sample.

Perfect samples yielded the lowest SAM values (≈ 2 – 3°), while Mask and Spot defects exhibited higher angles (≈ 4 – 6°).

Shape defects 17 and 22 exceeded the tolerance limit (3.2°), reflecting strong spectral-shape distortion.

Despite this variability between samples, inter-operator spread remained minimal, confirming SAM as a stable and operator-independent metric.

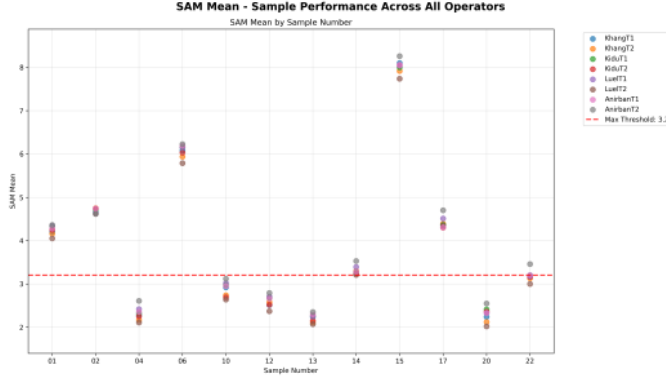


Fig. 11: SAM Across All Operators

CTQ-3: Uniformity Reproducibility Across Operators:

Uniformity measurements showed strong reproducibility across operators and trials.

Perfect samples displayed consistently high uniformity (0.8–0.9), while Homogeneity defects (6, 15) and Shape anomalies (17, 22) produced consistently low values (≈ 0.2 –0.3).

Mask and Spot defects remained above the acceptance threshold ($U \geq 0.4$).

The separation between uniform and non-uniform samples was sharp and operator-invariant, highlighting the robustness of the sector-based uniformity metric.

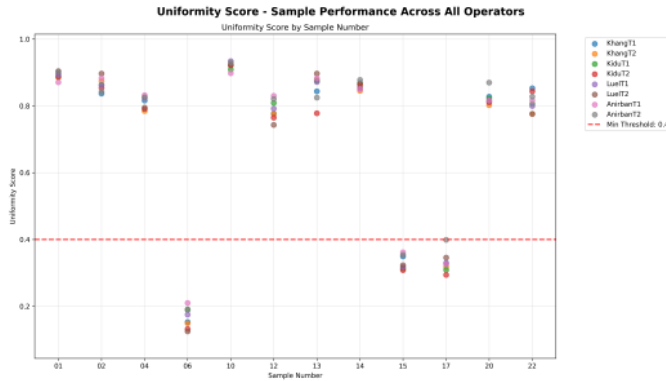


Fig. 12: Uniformity Across All Operators

Summary — Phase 2. Across all three CTQs, operator-to-operator and trial-to-trial variation remained extremely small. All CTQs produced stable and repeatable values across the full measurement workflow. This supports the conclusion that the hyperspectral measurement system satisfies industrial-grade measurement-system capability requirements.

C. Gage R&R Result Analysis

To evaluate the reliability of the hyperspectral measurement system, a crossed Gage Repeatability and Reproducibility (Gage R&R) study was performed using four operators, twelve IC grating samples, and two repeated measurements per operator. Analysis of Variance (ANOVA) was used to decompose the overall variability into repeatability (within-operator), reproducibility (between-operator), and part-to-part components. From these components, the total Gage variation, total study variation, percentage of Gage R&R, and the number of distinct categories (NDC) were computed to assess measurement-system capability. A summary of specification limits and input parameters is provided in Table III.

Input Summary	
Data Type	Continuous
Analysis Type	Crossed
Samples	12
Operators	4
Trials	2
Spec. Limits RMSE	[0, 0.08]
Spec. Limits SAM	[0, 3.2]
Spec. Limits Uniformity	[0.4, 1]

TABLE II: Summary of inputs used for Gage R&R evaluation.

Gage R&R Results for SRMSE: The Gage R&R results for SRMSE (Fig. 13) show that the measurement system contributes 11.68% of the total study variation and 25.59% of the allowable tolerance range. These values fall well within industry-accepted limits, indicating that SRMSE can be measured with adequate precision. All measurement error originates from repeatability, while reproducibility is effectively zero, demonstrating complete operator consistency. Part-to-part variation dominates (99.32%), confirming that SRMSE effectively differentiates among the twelve IC samples. The system achieves an NDC of 11, providing sufficient resolution for reliable discrimination. The detailed ANOVA and analysis summaries are provided in Tables III and IV.

ANOVA Table for RMSE					
Breakdown	DOF	SSQ	MS	F	P
Part	11	0.07	0.01	699.86	0.00
Operator	3	0.00	0.00	0.54	0.66
Part×Operator	33	0.00	0.00	0.83	0.71
Repeatability	48	0.00	0.00		
Total	95	0.08			

TABLE III: ANOVA results for RMSE.

Analysis (RMSE)	Results	Std Dev.	Study Var.	Tolerance
Total Gage R&R	0.00		11.68%	25.59%
Repeatability	0.00		11.68%	25.59%
Reproducibility	0.00		0.00%	0.00%
Part-to-Part	0.03		99.32%	217.57%
Total Variation	0.03		100.00%	219.07%
Number of Distinct Categories: 11				

TABLE IV: Gage R&R analysis results for RMSE.

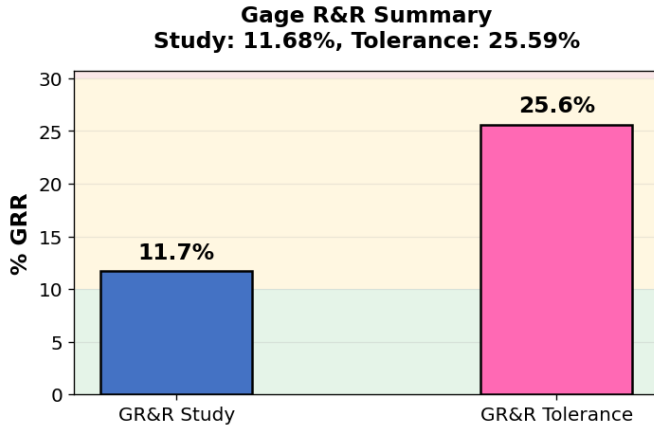


Fig. 13: Gage R&R Summary for RMSE.

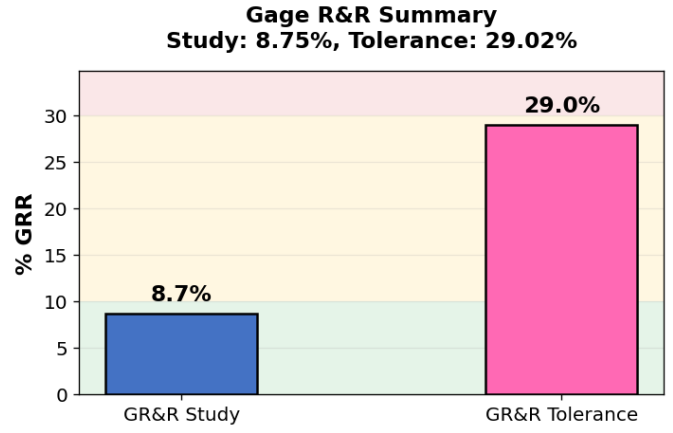


Fig. 14: Gage R&R Summary for SAM.

Gage R&R Results for SAM: The SAM results (Fig. 14) indicate a highly capable measurement system, with total Gage R&R contributing only 8.75% of the study variation. Repeatability (7.53%) and reproducibility (4.45%) remain well below acceptable limits, demonstrating minimal instrument and operator influence. Part-to-part variation (99.62%) overwhelmingly dominates the total variability, confirming strong differentiation capability. An NDC of 16 further demonstrates excellent measurement resolution. The ANOVA and summary tables for SAM are provided in Tables V and VI.

ANOVA Table for SAM					
Breakdown	DOF	SSQ	MS	F	P
Part	11	273.37	24.85	4855.04	0.00
Operator	3	0.46	0.15	30.02	0.00
Part×Operator	33	0.17	0.01	0.29	1.00
Repeatability	48	0.85	0.02		
Total	95	274.85			

TABLE V: ANOVA results for SAM.

Analysis Results (SAM)	Std Dev.	Study Var.	Tolerance
Total Gage R&R	0.15	8.75%	29.02%
Repeatability	0.13	7.53%	24.99%
Reproducibility	0.08	4.45%	14.75%
Part-to-Part	1.76	99.62%	330.44%
Total Variation	1.77	100.00%	331.71%

Number of Distinct Categories: 16

TABLE VI: Summary of Gage R&R analysis results for SAM.

Gage R&R Results for Uniformity: The Gage R&R analysis for Uniformity (Fig. 19) shows that the measurement system performs robustly, with total Gage R&R contributing only 9.61% of the study variation. Repeatability accounts for 9.05%, while reproducibility contributes 3.25%, indicating low instrument and operator influence. Part-to-part variation dominates at 99.54%, confirming that the metric reliably discriminates between homogeneous and non-homogeneous samples. An NDC of 14 demonstrates strong measurement resolution. Detailed ANOVA and analysis results are provided in Tables VII and VIII.

ANOVA Table for Uniformity					
Breakdown	DOF	SSQ	MS	F	P
Part	11	6.15	0.56	1190.22	0.00
Operator	3	0.01	0.00	4.80	0.01
Part×Operator	33	0.02	0.00	0.81	0.73
Repeatability	48	0.03	0.00		
Total	95	6.20			

TABLE VII: ANOVA results for Uniformity.

Analysis Results (Uniformity)	Std Dev.	Study Var.	Tolerance
Total Gage R&R	0.03	9.61%	25.51%
Repeatability	0.02	9.05%	24.01%
Reproducibility	0.01	3.25%	8.62%
Part-to-Part	0.26	99.54%	264.16%
Total Variation	0.27	100.00%	265.39%

Number of Distinct Categories: 14

TABLE VIII: Gage R&R analysis results for Uniformity.

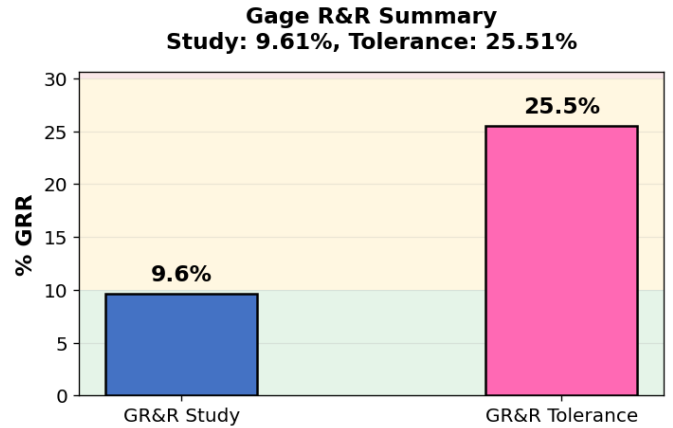


Fig. 15: Gage R&R Summary for Uniformity.

Overall Findings: Across all three CTQs, the Gage R&R results show that the proposed hyperspectral measurement system achieves high measurement reliability suitable for industrial deployment. Total Gage R&R values for SRMSE (11.68%), Uniformity (9.61%), and SAM (8.75%) are all

below the 30% tolerance limit, with SAM and Uniformity falling within the < 10% “excellent” category.

When analyzed relative to the specification limits ($\text{SRMSE} \in [0, 0.08]$, $\text{SAM} \in [0, 3.2^\circ]$, $\text{Uniformity} \in [0.4, 1.0]$), the percentage of Gage R&R with respect to tolerance (25.59%, 25.51%, and 29.02%) remains within acceptable bounds. Operator-induced variability is negligible across all CTQs, while part-to-part variation exceeds 99%, confirming that the CTQs reflect genuine physical differences among samples rather than measurement noise.

High NDC values (11 for SRMSE, 14 for Uniformity, and 16 for SAM) further demonstrate excellent resolution and discrimination capability. Collectively, these results validate that the hyperspectral acquisition, registration, and CTQ computation pipeline exhibits strong repeatability, reproducibility, and tolerance-conformant accuracy, satisfying Measurement System Analysis (MSA) criteria for industrial IC-grating quality control.

V. DISCUSSION

The results from both experimental phases demonstrate that the proposed hyperspectral measurement system provides a reliable and interpretable framework for IC grating quality assessment. Phase 1 confirmed that the three CTQs—SRMSE, SAM, and Uniformity—capture complementary aspects of the grating’s physical and optical behavior. SRMSE isolates magnitude and brightness-related deviations caused by fill-factor or structural variation, SAM highlights spectral-shape distortions associated with material or mask shifts, and Uniformity detects spatial gradients and sector-level inconsistencies. Their combined responses allowed clear separation of all defect categories, validating the CTQ definitions derived from customer requirements.

Phase 2 further established that the measurement pipeline is stable across operators and trials. The Gage R&R results showed that measurement-system variation contributes less than 12% of total study variation for all CTQs and less than 30% of the allowable tolerance, meeting industrial acceptance criteria. Operator influence was negligible, and part-to-part variation dominated the variance structure, confirming that observed differences arise from true physical defects rather than measurement artifacts. High NDC values (11–16) demonstrate strong discrimination capability. These findings collectively confirm that the system is both robust and suitable for high-volume manufacturing quality control.

VI. CONCLUSION

This work presents a complete hyperspectral metrology framework for automated inspection of diffractive in-coupler gratings used in AR waveguide displays. By integrating reflectance-based hyperspectral imaging, geometric alignment, and three customer-driven Critical-to-Quality (CTQ) metrics, the system provides objective, repeatable, and physically meaningful defect assessment. Phase 1 validated the discriminative capability of SRMSE, SAM, and Uniformity across all major defect categories, while Phase 2 confirmed the system’s

measurement reliability through a comprehensive Gage R&R study involving four operators and twelve representative samples.

All CTQs exhibited low measurement uncertainty, strong repeatability, and high resolution, demonstrating suitability for industrial deployment. The proposed methodology enables consistent classification of IC grating quality, supports improved process control, and offers a scalable foundation for future in-line inspection solutions. Future work may extend the system toward real-time acquisition, machine-learning-based defect prediction, or integration with manufacturing feedback loops to further enhance AR waveguide production yield.

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APPENDIX

APPENDIX A: SAMPLE MEASUREMENT ORDER

TABLE IX: Trial-1:Sample order of measurements for each operator

Order	Operator 1 (Anirban)	Operator 2 (Kidu)	Operator 3 (Khang)	Operator 4 (Luel)
1	Sample_ID0020	Sample_ID0001	Sample_ID0013	Sample_ID0022
2	Sample_ID0002	Sample_ID0002	Sample_ID0006	Sample_ID0014
3	Sample_ID0001	Sample_ID0006	Sample_ID0017	Sample_ID0010
4	Sample_ID0010	Sample_ID0017	Sample_ID0014	Sample_ID0020
5	Sample_ID0006	Sample_ID0010	Sample_ID0004	Sample_ID0001
6	Sample_ID0022	Sample_ID0012	Sample_ID0022	Sample_ID0013
7	Sample_ID0013	Sample_ID0013	Sample_ID0012	Sample_ID0004
8	Sample_ID0017	Sample_ID0014	Sample_ID0015	Sample_ID0012
9	Sample_ID0015	Sample_ID0015	Sample_ID0020	Sample_ID0006
10	Sample_ID0004	Sample_ID0020	Sample_ID0001	Sample_ID0015
11	Sample_ID0012	Sample_ID0022	Sample_ID0002	Sample_ID0002
12	Sample_ID0014	Sample_ID0004	Sample_ID0010	Sample_ID0017

TABLE X: Trial-2: Sample order of measurements for each operator

Order	Operator 1 (Anirban)	Operator 2 (Kidu)	Operator 3 (Khang)	Operator 4 (Luel)
1	Sample_ID0002	Sample_ID0017	Sample_ID0001	Sample_ID0017
2	Sample_ID0013	Sample_ID0001	Sample_ID0022	Sample_ID0004
3	Sample_ID0001	Sample_ID0006	Sample_ID0010	Sample_ID0020
4	Sample_ID0010	Sample_ID0020	Sample_ID0017	Sample_ID0010
5	Sample_ID0006	Sample_ID0004	Sample_ID0006	Sample_ID0022
6	Sample_ID0022	Sample_ID0010	Sample_ID0013	Sample_ID0015
7	Sample_ID0020	Sample_ID0012	Sample_ID0004	Sample_ID0013
8	Sample_ID0004	Sample_ID0013	Sample_ID0012	Sample_ID0001
9	Sample_ID0012	Sample_ID0002	Sample_ID0014	Sample_ID0002
10	Sample_ID0014	Sample_ID0015	Sample_ID0002	Sample_ID0014
11	Sample_ID0015	Sample_ID0022	Sample_ID0015	Sample_ID0012
12	Sample_ID0017	Sample_ID0014	Sample_ID0020	Sample_ID0006

Interaction Plot (Sample × Operator)
N: 12, YMin: 2.02, YMax: 8.26

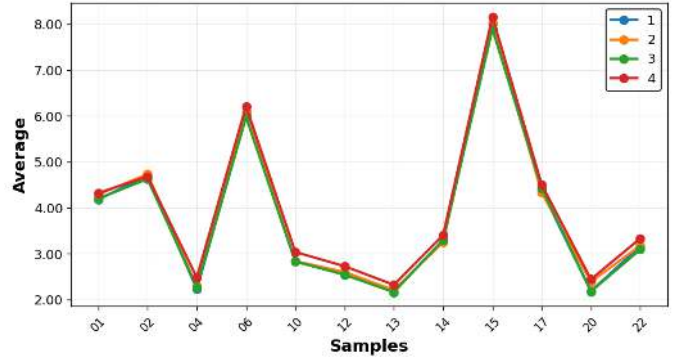


Fig. 18: Sample x Operator interaction for SAM

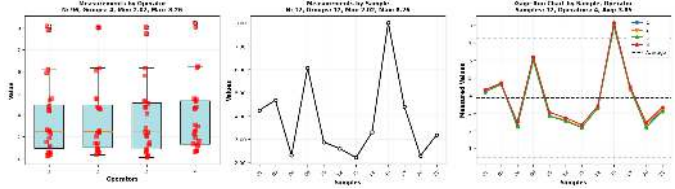


Fig. 19: Measurement by Operator(left), by sample(middle), Gage run chart by sample and operator(right) for SAM

Interaction Plot (Sample × Operator)
N: 12, YMin: 0.03, YMax: 0.12

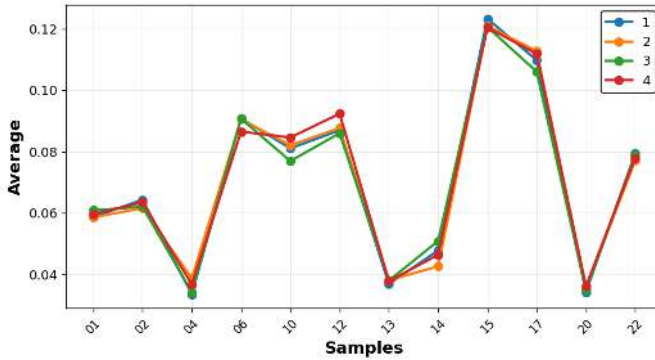


Fig. 16: Sample x Operator interaction for RMSE

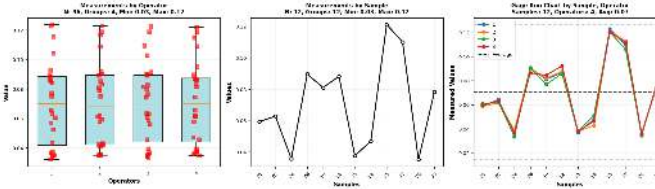


Fig. 17: Measurement by Operator(left), by sample(middle), Gage run chart by sample and operator(right) for RMSE

Interaction Plot (Sample × Operator)
N: 12, YMin: 0.12, YMax: 0.93

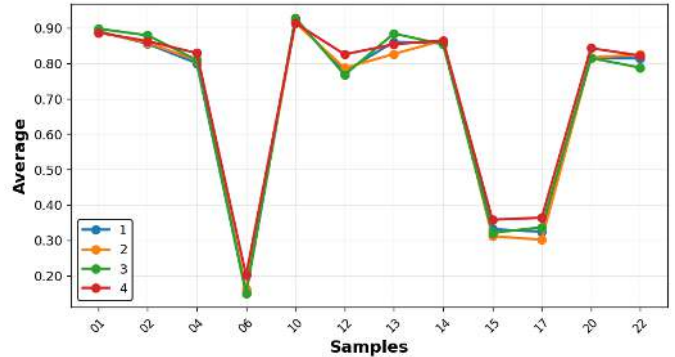


Fig. 20: Sample x Operator interaction for Uniformity

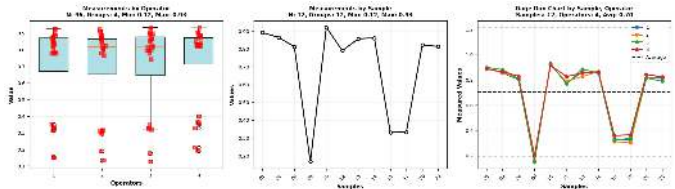


Fig. 21: Measurement by Operator(left), by sample(middle), Gage run chart by sample and operator(right) for Uniformity